Pre-Analysis Plan: #ManyDaughters Study Using SOEP-Core

1. Introduction and Research Hypotheses

This study investigates the causal impact of having daughters on various outcomes using data from the German Socio-Economic Panel (SOEP). We test four hypotheses:

H1: *Having daughters leads to stronger support for gender equality.*

H2: *Having daughters leads to more left-leaning political preferences.*

H3: *Having daughters leads to more gender equality in the household.*

H4: *Having daughters leads to a larger increase in the support of gender equality for men than for women.*

2. Data and Sample Construction

Dataset: SOEP-Core, covering the period 2000–2020 and including 90,357 individuals who completed the individual questionnaire in at least one of these years.

Unit of Analysis: Individual-level, with the number of daughters identified through data from individual questionnaires (dataset pl).

We include respondents who completed the individual questionnaire in at least one wave and who are parents with at least one biological child. Those without children or with missing data on key variables are excluded.

3. Variable Definitions and Coding

3.1. Main Independent Variable

The key independent variable, "Having_Daughters," is an indicator equal to 1 if the respondent has at least one biological daughter (pld0025, pld0026). Alternative definitions may include the number of daughters (pld0026, pld0027, pld0028).

3.2. Dependent Variables

H1 + H4: Support for Gender Equality (Attitudes)

Primary responses are taken from plh0395i01–plh0395i06, capturing attitudes towards genders. Robustness analysis may use indices constructed via factor analysis.

H2: Political Preference (Left-Leaning)

The primary measure is plh0004, which captures the respondent's self-placement on the left–right political spectrum (ranging from 0 = far left to 10 = far right). For the main analysis, we reverse the variable so that higher values reflect a more left-leaning orientation.

As robustness checks, we may also use: (i) a binary indicator equal to 1 if the original score is less than 5 (i.e., left-leaning), and 0 otherwise; and (ii) a party-based classification from plh0174_v1, where SPD, Greens, and Die Linke are coded as left-leaning.

H3: Gender Equality in the Household (Behavioral)

To measure gender equality in household labor, we construct a variable capturing the balance of domestic work between spouses, based on time-use reports. This variable is defined as follows:

$$Hausehold_Equality \;=\; 1 - \frac{|Husband's\ hours - Wife's\ hours|}{(Husband's\ hours \;+\; Wife's\ hours)}$$

where *Husband's hours and Wife's hours* refer to the reported hours spent on domestic tasks by each partner. The score ranges from 0 (complete inequality—one partner does all the work) to 1 (perfect equality—equal contributions).

4. Empirical Strategy

Each hypothesis is tested with the following regressions:

- To test H1: $Support_Gender_Equality_{it} = \beta_0 + \beta_1 * Having_Daughters_{it} +$ Controls $+ \varepsilon_{it}$

  where *Support_Gender_Equality* is one of six variables built based on answers from variables plh0395i01–plh0395i06. *Having_Daughters* is a dummy variable that equals one if the individual has at least one daughter, and zero otherwise.

- To test H2: $Political_Preference_Left_{it} = \beta_0 + \beta_1 * {}_{it} Having_Daughters +$ Controls $+ \varepsilon_{it}$

  Political_Preference_Left is measured with plh0004, which captures the respondent's self-placement on the left–right political spectrum (ranging from 0 = far left to 10 = far right). For the main analysis, we reverse the variable so that higher values reflect a more left-leaning orientation.

- To test H3: $Hausehold_Equality_{it} = \beta_0 + \beta_1 * Having_Daughters_{it} + Controls + \varepsilon_{it}$

  EqualityScore is a variable reflecting the distribution of domestic labor, as outlined in Section 4.

- To test H4: $Support_Gender_Equality_{it} = \beta_0 + \beta_1 * Having_Daughters_{it} + \beta_2 * Gender_i + \beta_3 * Having_Daughters_{it} * Gender_i + Controls + \varepsilon_{it}$

Controls includes a set of critical individual and household control variables, such as:

- Age of the respondent
- Gender and marital status of the respondent
- Household size
- Employment status and working hours
- Income, public transfers, and wealth proxies
- Education level and parental education
- Number of children
- Number of years since having the first child
- Migration background and region
- Self-rated health status
- Religion of the respondent

All regressions include year fixed effects, and standard errors are clustered at the household level. Additional models may explore household or cohort fixed effects.

6. Interpretation and Reporting

Each model will report:

- Coefficient and standard error
- Test statistics (e.g., t-value)
- F-statistic with degrees of freedom

- P-value
- Number of observations and unique respondents
- Interpretation of effect direction

Only the results of primary models are exported to the official CSVs. Supplementary checks remain interpretive.

7. Deliverables and Implementation Notes

- Each hypothesis is implemented in a separate do-file.
- Results are exported as CSV files with standard format.
- Code is written in Stata 14 or later.
- Supplementary results stored separately.

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